# HDU Spring Round II 2026

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Mountain of Faith Team

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# Problem A. Afternoon by the Lake

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 8 s |
| Memory limit: | 512 MB |

Cirno gives you two non-negative integers $s$ and $t$ in binary form. Determine how many pairs $(a, b)$ satisfy

$$a \times b = (a \texttt{ or } b) \times (a \texttt{ and } b)$$

where both $a$ and $b$ are integers in the interval $[s, t]$.

Here, `or` denotes the bitwise OR operation, and `and` denotes the bitwise AND operation.

Since the answer may be very large, output it modulo 998244353.

## Input

The first line contains an integer $Q$ ($1 \le Q \le 20$), the number of test cases.

Each test case consists of one line containing two binary strings $S$ and $T$, representing the binary forms of $s$ and $t$, respectively.

It is guaranteed that:

$$1 \le |S|, |T| \le 5 \times 10^5,$$

the binary representations of $s$ and $t$ have no leading zeros, and $s \le t$.

## Output

For each test case, output one line containing a non-negative integer: the number of valid pairs modulo 998244353.

## Examples

| standard input | standard output |
|---|---|
| 6<br>11 1000<br>1000 1001<br>0 100<br>11 111<br>10 111<br>0 11111 | 18<br>4<br>17<br>17<br>24<br>454 |

## Note

In the equation, `or` means bitwise OR, and `and` means bitwise AND.

# Problem B. Blossoms of Cherry

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Youmu's garden can be regarded as an electronic maze on the Cartesian plane. There are $n$ laser emitters in the first quadrant. The $i$-th emitter is located at $(x_i, y_i)$, and it emits a laser in exactly one of the four directions: up, right, down, or left.

We say that a point lies on the illumination path of an emitter if and only if it lies on the ray that starts from the emitter and extends in its shooting direction. Note that the emitter itself is also on its own illumination path.

Because the lasers are very powerful, if two emitters illuminate each other simultaneously, it would be dangerous. Therefore, it is guaranteed that **there do not exist two emitters such that each lies on the other's illumination path**.

To prevent light pollution, Youmu wants to place some obstacles in the electronic maze (an obstacle may also be placed at the position of an emitter) so that no laser can extend to infinity. She wants to know the minimum number of obstacles required so that every emitter's illumination path contains at least one obstacle.

## Input

The first line contains an integer $t$ ($1 \le t \le 100$), the number of test cases.

Each test case begins with an integer $n$ ($1 \le n \le 300$), the number of laser emitters.

Then follow $n$ lines. The $i$-th of them contains three integers $x_i$, $y_i$, and $d_i$ ($1 \le x_i, y_i \le 10^9$, $d_i \in \{0, 1, 2, 3\}$), meaning that the $i$-th emitter is located at $(x_i, y_i)$, and:

- if $d_i = 0$, it emits upward;
- if $d_i = 1$, it emits to the right;
- if $d_i = 2$, it emits downward;
- if $d_i = 3$, it emits to the left.

It is guaranteed that the sum of $n$ over all test cases does not exceed 3000.

## Output

For each test case, output one line containing a single integer: the minimum number of obstacles that Youmu needs to place.

## Examples

| standard input | standard output |
|---|---|
| 2<br>6<br>5 5 0<br>3 7 1<br>9 9 2<br>9 1 3<br>5 2 0<br>1 20 3<br>1<br>1 1 0 | 3<br>1 |

## Note

In the first sample test case, one possible solution is to place obstacles at $(5, 7)$, $(9, 1)$, and $(-2025, 20)$.

The obstacle at $(5, 7)$ blocks the 1st, 2nd, and 5th lasers; the obstacle at $(9, 1)$ blocks the 3rd and 4th lasers; and the obstacle at $(-2025, 20)$ blocks the 6th laser.

It can be proven that no better solution exists.

# Problem C. Calm Melody of Bamboo

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 6 s |
| Memory limit: | 512 MB |

You are given a sequence of length $n$: $a_1, a_2, \ldots, a_n$, and a fixed value $x$.

Reisen and Tewi take turns choosing numbers from the sequence. Whenever a number is chosen, it is deleted immediately.

Reisen moves first. Then they alternate until the sequence becomes empty.

The rules are as follows:

- On each of her turns, Reisen may choose any one number in the current sequence.
- On each of her turns, Tewi must take the first $x$ numbers of the current sequence, namely $a_1, a_2, \ldots, a_x$. If the current sequence contains fewer than $x$ numbers, then she takes all of them.

Reisen wants to maximize the sum of the numbers she takes. Suppose Reisen takes $a_{k_1}, a_{k_2}, \ldots, a_{k_m}$; then she wants to maximize

$$\sum_{i=1}^{m} a_{k_i}.$$

Assuming Reisen always plays optimally, for every $x = 1, 2, 3, \ldots, n-1$, independently compute the maximum possible sum of the numbers taken by Reisen.

## Input

The first line contains an integer $t$ ($1 \le t \le 1000$), the number of test cases.

Each test case begins with an integer $n$ ($2 \le n \le 2 \times 10^5$), the length of the sequence.

The second line of each test case contains $n$ positive integers $a_i$ ($1 \le a_i \le 10^9$), describing the sequence.

It is guaranteed that the sum of $n$ over all test cases does not exceed $5 \times 10^5$.

## Output

For each test case, output one line containing $n-1$ integers. The $i$-th integer should be the maximum possible sum of the numbers taken by Reisen when $x = i$.

## Examples

| standard input | standard output |
|---|---|
| 3<br>2<br>1 2<br>6<br>1 1 4 5 1 4<br>11<br>12 11 10 9 1 2 3 4 5 12 11 | 2<br>13 9 9 9 5<br>54 44 35 35 24 24 24 24 23 12 |

## Note

In the second sample test case, the initial sequence is $1, 1, 4, 5, 1, 4$.

When $x = 1$, one optimal strategy for Reisen is:

- Reisen first takes $a_4 = 5$. The sequence becomes $1, 1, 4, 1, 4$.
- Tewi must then take the first 1 element, namely $a_1 = 1$. The sequence becomes $1, 4, 1, 4$.
- Reisen then takes $a_2 = 4$. The sequence becomes $1, 1, 4$.
- Tewi must take the first 1 element, namely $a_1 = 1$. The sequence becomes $1, 4$.
- Reisen then takes $a_2 = 4$. The sequence becomes 1.
- Tewi must take the first 1 element, namely $a_1 = 1$. The sequence becomes empty, and the game ends.

The sum of the numbers taken by Reisen is $5 + 4 + 4 = 13$. It can be proven that this is optimal.

When $x = 3$, one optimal strategy for Reisen is:

- Reisen first takes $a_4 = 5$. The sequence becomes $1, 1, 4, 1, 4$.
- Tewi must then take the first 3 elements, namely $a_1 = 1$, $a_2 = 1$, and $a_3 = 4$. The sequence becomes $1, 4$.
- Reisen then takes $a_2 = 4$. The sequence becomes 1.
- Tewi must take the first 3 elements, but since fewer than 3 elements remain, she takes all of them. The sequence becomes empty, and the game ends.

The sum of the numbers taken by Reisen is $5 + 4 = 9$. It can be proven that this is optimal.

# Problem D. Down by the Sanzu Riverbank

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Given two non-negative integers $n$ and $k$, Shikieiki asks you to construct $n$ positive integers such that their product is greater than their sum, and the difference is exactly $k$.

In other words, you need to find $n$ positive integers $x_1, x_2, \ldots, x_n$ satisfying

$$(x_1 + x_2 + \cdots + x_n) + k = x_1 \times x_2 \times \cdots \times x_n.$$

Since the output numbers must not be too large, you also need to guarantee that for every constructed number $x_i$,

$$1 \le x_i \le 10^8.$$

If there are multiple solutions, output any of them.

## Input

The first line contains an integer $t$ ($1 \le t \le 3000$), the number of test cases.

Each test case consists of one line containing two integers $n$ and $k$ ($2 \le n \le 2 \times 10^5$, $0 \le k \le 10^7$).

It is guaranteed that the sum of $n$ over all test cases does not exceed $5 \times 10^5$.

## Output

For each test case, output one line containing $n$ positive integers $x_1, x_2, \ldots, x_n$ ($1 \le x_i \le 10^8$).

It can be proven that under the given constraints, a solution always exists.

If there are multiple solutions, output any one of them.

## Examples

| standard input | standard output |
|---|---|
| 3<br>3 90<br>3 970<br>2 2 | 3 5 7<br>10 10 10<br>4 2 |

## Note

In the first sample test case, $k = 90$, and $[3, 5, 7]$ is a valid solution because

$$(3 \times 5 \times 7) - (3 + 5 + 7) = 90 = k.$$

In the second sample test case, $k = 970$, and $[10, 10, 10]$ is a valid solution because

$$(10 \times 10 \times 10) - (10 + 10 + 10) = 970 = k.$$

In the third sample test case, $k = 2$, and $[4, 2]$ is a valid solution because

$$(4 \times 2) - (4 + 2) = 2 = k.$$

# Problem E. Early Breeze at Moriya Shrine

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 14 s |
| Memory limit: | 512 MB |

Sanae are seeking your help for Kanako's mission!

You are given $n$ circular regions. The center of the $i$-th circle is $(p_i, q_i)$, and its radius is $r_i$. The corresponding circular region is

$$\{(x, y) \mid (x - p_i)^2 + (y - q_i)^2 \le r_i^2\}.$$

It is guaranteed that the radius of every circle is at most 30.

You are also given $m$ queries. In each query, four integers $x_{i,1}$, $y_{i,1}$, $x_{i,2}$, $y_{i,2}$ are given, where $x_{i,1} \le x_{i,2}$ and $y_{i,1} \le y_{i,2}$. They describe the rectangular region whose four vertices are

$$(x_{i,1}, y_{i,1}),\ (x_{i,1}, y_{i,2}),\ (x_{i,2}, y_{i,2}),\ (x_{i,2}, y_{i,1}),$$

that is,

$$\{(x, y) \mid x \in [x_{i,1}, x_{i,2}],\ y \in [y_{i,1}, y_{i,2}]\}.$$

For each query, determine how many of the given $n$ circular regions have at least one common point with the given rectangular region.

## Input

The first line contains an integer $t$ ($1 \le t \le 1000$), the number of test cases.

Each test case begins with two integers $n$ and $m$ ($1 \le n \le 1.5 \times 10^6$, $1 \le m \le 10^4$), denoting the number of circles and the number of queries, respectively.

Then follow $n$ lines. The $i$-th of them contains three integers $p_i$, $q_i$, and $r_i$ ($1 \le p_i, q_i \le 10^9$, $1 \le r_i \le 30$), meaning that the center of the $i$-th circular region is $(p_i, q_i)$ and its radius is $r_i$.

Then follow $m$ lines. The $i$-th of them contains four integers $x_{i,1}$, $y_{i,1}$, $x_{i,2}$, and $y_{i,2}$ ($1 \le x_{i,1} \le x_{i,2} \le 10^9$, $1 \le y_{i,1} \le y_{i,2} \le 10^9$), describing the queried rectangular region.

It is guaranteed that the sum of $n$ over all test cases does not exceed $3 \times 10^6$, and the sum of $m$ over all test cases does not exceed $2 \times 10^4$.

## Output

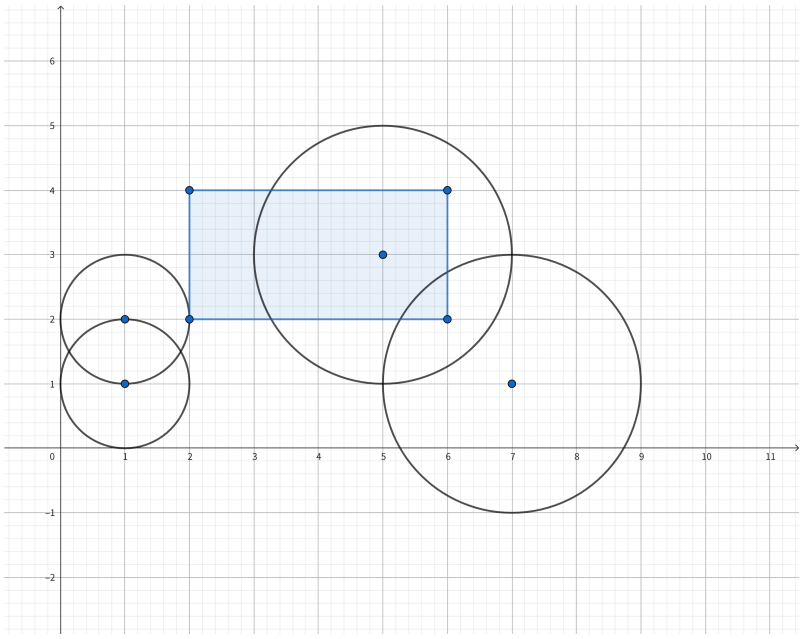
For each test case, output $m$ lines. On the $i$-th line, output a non-negative integer: the answer to the $i$-th query.

## Examples

| standard input | standard output |
|---|---|
| 2<br>4 1<br>1 1 1<br>7 1 2<br>5 3 2<br>1 2 1<br>2 2 6 4<br>8 3<br>5 6 4<br>7 7 4<br>3 3 2<br>1 8 5<br>10 1 2<br>1 8 1<br>7 2 2<br>5 4 5<br>9 1 10 5<br>3 1 6 3<br>5 1 9 5 | 3<br>4<br>4<br>7 |

## Note

In the first sample test case, the queried rectangle has common points with the 2nd, 3rd, and 4th circles, so the answer is 3.



# Problem F. Family Reading

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Satori gives Koishi an array $a_1, a_2, \ldots, a_n$ of length $n$, and you are called to help Koishi to determine the maximum possible number of indices $i$ $(1 \le i \le n)$ such that

$$a_i = i$$

after rearranging the array.

Here, a rearrangement means permuting the elements of the original array $a_1, a_2, \ldots, a_n$ in any order to obtain a new array. The new array must contain exactly the same elements as the original one, including duplicates.

## Input

The first line contains an integer $t$ $(1 \le t \le 10^5)$, the number of test cases.

Each test case consists of two lines.

The first line contains a positive integer $n$ $(1 \le n \le 5 \times 10^5)$, the length of the array.

The second line contains $n$ positive integers, where the $i$-th integer is $a_i$ $(1 \le a_i \le 10^9)$.

It is guaranteed that the sum of all $n$ over all test cases does not exceed $2 \times 10^6$.

## Output

For each test case, output one line containing a non-negative integer: the maximum possible number of indices $i$ such that $a_i = i$ after rearranging the array.

## Examples

| standard input | standard output |
|---|---|
| 2<br>5<br>5 3 2 1 3<br>6<br>4 2 1 9 8 7 | 4<br>3 |

## Note

In the first sample test case, one possible rearrangement is $[1, 2, 3, 3, 5]$. Then positions $1, 2, 3, 5$ satisfy $a_i = i$. It can be proven that the maximum possible answer is 4.

In the second sample test case, one possible rearrangement is $[1, 2, 8, 4, 7, 9]$. It can be proven that the maximum possible answer is 3.

# Problem G. Gliding Above the Clouds

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

For a multiset $S = \{a_1, a_2, \cdots, a_k\}$ of $k$ non-negative integers, define its value as

$$a_1 \oplus a_2 \oplus \cdots \oplus a_k,$$

where $\oplus$ denotes bitwise XOR. In particular, the value of the empty set $\varnothing$ is defined to be 0.

Toramaru Shou and Nazrin play a game on a multiset $S = \{a_1, a_2, \cdots, a_k\}$. Shou moves first, and the two players alternate turns.

Suppose the current multiset is $S$. A move consists of:

- choosing an element $a_i \in S$ and deleting it from $S$.

That is, if $S = \{a_1, a_2, \cdots, a_m\}$, then after deleting $a_i$, the multiset becomes

$$S \leftarrow \{a_1, a_2, \cdots, a_{i-1}, a_{i+1}, a_{i+2}, \cdots, a_m\}.$$

At any moment, if the value of $S$ becomes 0, then the game stops immediately, and the player who made the last move loses.

If nobody has made a move yet, i.e. the initial value of $S$ is already 0, then Shou loses.

You are given an array of length $n$, $x_1, x_2, \cdots, x_n$. For every subarray $[l, r]$ $(1 \le l \le r \le n)$, Shou and Nazrin play one game on the multiset

$$S = \{x_l, x_{l+1}, \cdots, x_r\}.$$

Assuming both players play optimally, determine the number of subarrays on which Shou wins and the number of subarrays on which Nazrin wins.

## Input

The first line contains an integer $t$ $(1 \le t \le 100)$, the number of test cases.

Each test case consists of two lines.

The first line contains an integer $n$ $(1 \le n \le 2 \times 10^5)$.

The second line contains $n$ non-negative integers $x_1, x_2, \cdots, x_n$ $(0 \le x_i \le 2 \times 10^5)$.

It is guaranteed that the sum of all $n$ over all test cases does not exceed $5 \times 10^5$.

## Output

For each test case, output one line containing two integers: the number of subarrays on which Toramaru Shou wins, and the number of subarrays on which Nazrin wins.

## Examples

| standard input | standard output |
|---|---|
| 5<br>1<br>0<br>3<br>5 5 2<br>4<br>2 0 1 3<br>6<br>0 1 2 0 1 2<br>13<br>1 1 4 5 1 4 1 9 1 9 8 1 0 | 0 1<br>1 5<br>3 7<br>8 13<br>39 52 |

## Note

In the second sample test case, consider the game played on the multiset $S = \{5, 5\}$ corresponding to subarray $[1, 2]$. Its initial value is

$$5 \oplus 5 = 0,$$

so the game ends immediately before anyone makes a move. Therefore, Shou loses and Nazrin wins.

For the subarray $[2, 3]$, the multiset is $S = \{5, 2\}$, whose initial value is

$$5 \oplus 2 = 7.$$

Shou may delete 2, after which $S = \{5\}$ and its value is 5. Then Nazrin has no choice but to delete 5, after which $S = \varnothing$ and its value becomes 0. The game stops immediately, and since Nazrin made the last move, Nazrin loses. Hence Shou wins.

Therefore, in the second sample test case, Shou wins on subarray $[2, 3]$, while Nazrin wins on subarrays $[1, 1]$, $[1, 2]$, $[1, 3]$, $[2, 2]$, and $[3, 3]$.

# Problem H. Hearts of Ancient Capital

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Mononobe no Futo gives you $n$ integers $a_1, a_2, \ldots, a_n$, arranged in clockwise order on a circle. It is guaranteed that their sum is 0.

Soga no Tojiko wants to choose a starting position $k$, and then take $n$ consecutive numbers in clockwise order starting from $a_k$. She wants every prefix sum of this length-$n$ sequence to be non-negative.

It can be proven that such a $k$ always exists. If there are multiple valid choices, output the smallest one.

In other words, you need to find the smallest $k$ $(1 \le k \le n)$ such that for every $l$ $(0 \le l \le n-1)$,

$$\sum_{i=0}^{l} a_{k+i} \ge 0,$$

where

$$a_{n+1} = a_1,\ a_{n+2} = a_2,\ \ldots,\ a_{2n-1} = a_{n-1},$$

and it is guaranteed that

$$\sum_{i=1}^{n} a_i = 0.$$

## Input

The first line contains an integer $t$ $(1 \le t \le 3000)$, the number of test cases.

Each test case consists of two lines.

The first line contains an integer $n$ $(1 \le n \le 2 \times 10^5)$.

The second line contains $n$ integers $a_1, a_2, \cdots, a_n$ $(-10^9 \le a_i \le 10^9)$, representing the elements of the array.

It is guaranteed that the sum of $n$ over all test cases does not exceed $5 \times 10^5$.

## Output

For each test case, output one integer $k$ satisfying the condition in the statement.

## Examples

| standard input | standard output |
|---|---|
| 2<br>5<br>3 -1 -2 -2 2<br>2<br>1 -1 | 5<br>1 |

## Note

In the first sample test case, $k = 5$ satisfies the condition, because

$$a_5 = 2, \quad a_5 + a_6 = 5, \quad a_5 + a_6 + a_7 = 4, \quad a_5 + a_6 + a_7 + a_8 = 2, \quad a_5 + a_6 + a_7 + a_8 + a_9 = 0$$

are all non-negative.

It can be proven that there is no valid index smaller than 5.

# Problem I. In the Miniature Garden

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Shinmyoumaru Sukuna is playing a game called "Your Craft". In this game, she can freely create and destroy different kinds of blocks in a three-dimensional world filled with blocks.

Now there is a grid of $N \times M$ cells beneath Shinmyoumaru's feet. At first, she covers all these cells with white carpets, and then goes to sleep.

Unfortunately, while Shinmyoumaru is asleep, you want to make some small modifications to her carpets. You want to recolor some of the white carpets into black carpets, while satisfying the following two conditions:

- No two cells containing black carpets may share a common edge. In other words: if the cell in row $i$ and column $j$ is black, then among the cells $(i-1, j)$, $(i+1, j)$, $(i, j-1)$, and $(i, j+1)$, any that exist cannot be black.
- The final pattern must have at least one horizontal or vertical axis of symmetry, that is, at least one of the following two statements must hold:
  - For every $i \in [1, N]$ and $j \in [1, M]$, the color of the cell in row $i$ and column $j$ is the same as the color of the cell in row $N - i + 1$ and column $j$.
  - For every $i \in [1, N]$ and $j \in [1, M]$, the color of the cell in row $i$ and column $j$ is the same as the color of the cell in row $i$ and column $M - j + 1$.

Your task is to determine the maximum possible number of black carpets.

## Input

The first line contains an integer $t$ ($1 \le t \le 2 \times 10^4$), the number of test cases.

Each test case contains one line with two integers $N$ and $M$ ($1 \le N, M \le 10^9$).

## Output

For each test case, output one integer on a separate line, denoting the maximum possible number of black carpets.

## Examples

| standard input | standard output |
|---|---|
| 5<br>1 1<br>2 2<br>3 3<br>4 4<br>5 5 | 1<br>0<br>5<br>4<br>13 |

# Problem J. Jewels from the Silent Sea

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 s |
| Memory limit: | 512 MB |

Junko gives you two integers $x$ and $k$. Your task is to find the $k$-th sequence in lexicographical order among all non-empty sequences of positive integers whose sum of elements does not exceed $x$.

Suppose sequence $a$ has length $n$, and sequence $b$ has length $m$. We say that $a$ is lexicographically smaller than $b$ if and only if one of the following holds:

1. $n < m$, and for all $1 \le i \le n$, we have $a_i = b_i$;
2. there exists an integer $p$ ($1 \le p \le \min\{n, m\}$) such that for all $1 \le i < p$, we have $a_i = b_i$, and $a_p < b_p$.

## Input

The first line contains an integer $t$ ($1 \le t \le 10^5$), the number of test cases.

Each test case consists of one line containing two integers $x$ and $k$ ($1 \le x \le 10^6$, $1 \le k \le 10^{18}$), describing the query among all non-empty sequences of positive integers whose sum of elements does not exceed $x$.

It is guaranteed that the sequence with lexicographical rank $k$ always exists.

It is guaranteed that the sum of all $x$ over all test cases does not exceed $2 \times 10^6$.

## Output

For each test case, output one line containing several positive integers, representing the answer sequence.

Separate adjacent integers in the same line by a single space.

## Examples

| standard input | standard output |
|---|---|
| 3<br>1 1<br>3 3<br>4 10 | 1<br>1 1 1<br>2 1 |

## Note

In the second sample test case, the sequences in lexicographical order are:

$$[1],\ [1,1],\ [1,1,1],\ [1,2],\ [2],\ [2,1],\ [3].$$